KAUSHAL REDDY PARVATALA

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PROFILE SUMMARY

Computer Science Engineering student at Anurag University with hands-on experience in Python, JavaScript, REST APIs, database integration, and AI-powered applications. Built a 4-agent AI workflow and an interactive 3D motorsport visualization platform, with experience in modular architecture, automated testing, and LLM integration. Top 10 finisher at Techxcelerate (CodeBeats) among 100+ teams.

EDUCATION

B.Tech – Computer Science Engineering Expected July 2027
Anurag University Hyderabad
CGPA: 8.52

HACKATHONS & LEADERSHIP

- Led development of an AI-driven application at Techxcelerate (CodeBeats), securing a **Top 10 finish among 100+ teams** within **24 hours**.
- Coordinated task allocation, technical decisions, and feature integration across **4 national-level hackathons**, helping teams deliver functional applications under tight deadlines.
- Optimized application architecture across competition projects, achieving **30% faster data retrieval** and **30% lower bandwidth consumption**.

PROJECTS

Grant Agent System | AI Research Grant Proposal Generator & Evaluator
Python, CrewAI, Google Gemini API, SQLite, Streamlit
- Architected a **4-agent AI workflow** comprising Drafter, Planner, Evaluator, and Refiner agents, with explicit interfaces supporting modular development and testing.
- Established a deterministic validation layer for LLM-generated outputs, enforcing structured-output requirements and improving workflow consistency.
- Authored **15 unit tests** for the data-ingestion pipeline, identifying **2 timestamp-parsing defects** before dashboard integration.

APEX-F1 | Immersive Formula 1 Visualization Platform
Next.js, React, TypeScript, React Three Fiber, Three.js, OpenF1
- Built an interactive **3D Monaco circuit visualization** using cached OpenF1-derived geometry, validating **1,634 track points across 2 clean laps**.
- Engineered interactive 3D track rendering and camera controls to enable exploration of circuit geometry.
- Extended a modular, contract-oriented platform architecture to support future race-data integration and live visualization, with **11 repository commits**.

ReConnect – Alumni Networking Platform (In Progress)
Next.js, TypeScript, Tailwind CSS, Firebase Authentication, Cloud Firestore, Firebase Security Rules
- Designed a role-based platform supporting **3 user roles**—students, alumni, and administrators—with dedicated profile and dashboard workflows.
- Modeled application data across **6 Firestore collections** covering users, alumni profiles, mentorship requests, engagement logs, events, and opportunities.
- Developed mentorship workflows spanning **3 operations**: request submission, incoming-request retrieval, and accept/reject responses, supported by Firebase Authentication and Firestore security rules.

TECHNICAL SKILLS

Languages — Python, Java, C, C++, JavaScript
Backend & APIs — REST API Design & Integration, Firebase
Databases — MySQL, MongoDB, Cloud Firestore
CS Fundamentals — Data Structures & Algorithms, OOP, DBMS, Computer Networks
Dev Tools — Git, GitHub, VS Code, Postman, Linux, Vercel
AI & LLMs — Prompt Engineering, LLM API Integration, CrewAI, Streamlit